



# The clinical Alzheimer's disease spectrum classified in the A/T/N framework with $^{18}\text{F}$ -Flutemetamol-Centiloid, $^{18}\text{F}$ -MK-6240-CenTauR, and $^{18}\text{F}$ -FDG-AD metaROI: a multicenter memory clinic observational study

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## Abstract

**Purpose** This study pioneers the ability of semi-quantitative A/T/N classification using combined  $^{18}\text{F}$ -Flutemetamol (A),  $^{18}\text{F}$ -MK-6240 (T) and  $^{18}\text{F}$ -FDG (N) positron emission tomography (PET) tracers with binary classifications and standardised scales for Centiloid (A), CenTauR (T), and AD metaROI (N), to clinically describe subjects with Mild Cognitive Impairment (MCI) or Mild Dementia (MD) in Alzheimer's disease (AD).

**Methods** MCI ( $n=45$ , mean age  $71.0 \pm 8.8$ , 40% female, MMSE 26.2) and MD ( $n=44$ , mean age  $70.1 \pm 7.6$ , 48% female, MMSE 24.1) participants underwent PET with  $^{18}\text{F}$ -Flutemetamol for amyloid- $\beta$ ,  $^{18}\text{F}$ -MK-6240 tau and/or  $^{18}\text{F}$ -FDG for neurodegeneration. Images were semi-quantified using Centiloid, CenTauR, and AD metaROI. Predefined cut-offs classified scans as positive (+) or negative (-), and standard uptake value ratio (SUVR) values semi-quantified A-burden, T-burden, and N-burden.

**Results** T+ was more frequent in MD (68%) than in MCI (44%). Among T+, MD participants were more often A+ (39%) than MCI (27%). T-burden differed significantly between MCI and MD ( $W=642$ ,  $p=0.004$ ). Increasing A-burden, T-burden, and hypometabolism, reflected by lower N-burden were associated with worse performance in Norwegian validated revised mini-mental state-examination (MMSE-NR3) ( $p=0.048$ ,  $<0.001$ , and  $0.006$ , respectively), A-burden and N-burden with CERAD Word List Memory Test for immediate and delayed recall (CERAD) ( $p=0.041$ ), T-burden with CERAD Word List Memory Test for recognition (CERAD R) ( $p<0.001$ ), and N-burden with Clock-drawing test (CDT) ( $p=0.023$ ) and Controlled Oral Word Association Test (COWAT) ( $p=0.025$ ).

**Conclusion** The A/T/N classification was most informative for A (+/-) and T (+/-) PET, with T-PET appearing to distinguish MCI from MD, while N-PET correlated most with cognitive impairment.

**Keywords** Neuroimaging · Biomarkers · Vizamyl · Neurodegenerative Diseases · Tauopathy · Brain Metabolism

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## Introduction

The A/T/N framework is a classification scheme to assess the pathological processes across the Alzheimer's disease (AD) spectrum, ranging from preclinical to mild cognitive impairment (MCI) to dementia [1–3]. Here, A denotes the presence of amyloid- $\beta$  (A $\beta$ ), T tau deposits in the brain, and N represents neurodegeneration [2, 3]. The binary A/T/N classification stratifies individuals as positive (+) or negative (-) for A, T or N based on positron emission tomography (PET), cerebrospinal fluid (CSF), or, more recently, blood biomarkers [1, 2]. Using PET, the amount and spatial spread of A, T, and N can be visualised and quantified using specific tracers. For A,  $^{18}\text{F}$ -Flutemetamol has high sensitivity and specificity for detecting A $\beta$  in AD [4]. For T, the second-generation tau tracer  $^{18}\text{F}$ -MK-6240 has high sensitivity and specificity for early neurofibrillary tau (NFT) accumulation in AD and is not sensitive to non-AD tauopathies such as progressive supranuclear palsy and corticobasal degeneration [5, 6]. For N,  $^{18}\text{F}$ -Fluorodeoxyglucose (FDG), a well-established glucose metabolism tracer, can assess hypometabolism as a proxy of reduced synaptic activity.  $^{18}\text{F}$ -FDG is accurate in early and differential diagnosis of patterns in the AD spectrum [2]. There is a notable lack of studies combining multiple PET biomarkers within the same individual. One study combining A-PET and T-PET reported that adding T-PET aids the diagnosis of MCI and mild dementia (MD) within the AD spectrum [7]. Moreover, combining T-PET with N-PET could improve clinical diagnosis, disease evaluation, and management [8].

Existing studies have failed to adequately address the optimal combination and analysis of A/T/N PET scans to clinically distinguish MCI from MD. Studies using different tracer combinations are not directly comparable, in addition to differences in populations and analysis methods [9, 10]. Recent efforts have developed A, T, and N scales to standardise PET analysis, with recommended thresholds to assess PET scans as positive or negative. The most recognised are the Centiloid scale for A-PET [11], the CenTauR scale for T-PET [12], and the AD-specific meta-region-of-interest (AD metaROI) for N-PET [13, 14]. These scales are new and have rarely been used together [14–19]. The current study addressed this gap by systematically applying binary assessment of A-PET, T-PET, and N-PET using these standardised scales and cut-offs, to evaluate if such standardised PET interpretation, could improve MCI-MD differential diagnosis in the AD spectrum. MCI and MD are staged clinically based on cognitive and functional tests, questionnaires, clinical symptoms, and A $\beta$  and Tau pathology in CSF, PET, and/or blood [2, 10]. A fundamental weakness of these tests is their inability to identify and

distinguish pathology related to AD [2, 10]. Combinations of PET tracers for diagnosis and follow up of AD could support precision medicine as anti-A $\beta$  drugs are in clinical use and anti-tau treatments are in development.

Ideally, PET biomarkers should mirror clinical symptoms and AD stages. Previous studies showed that  $^{18}\text{F}$ -FDG hypometabolism correlates most with cognitive impairment, followed by tau, whereas A $\beta$  is least correlated with cognition, especially in advanced stages [13, 17, 19–21]. However, most of these studies used only one or two A/T/N PET tracers with standardised scales, three tracers without standardisation, or a single cognitive test (e.g. mini-mental state-examination (MMSE) or clinical dementia rating (CDR)) [13, 17, 19–21]. To address these limitations, we included the three A/T/N PET tracers, applied standardised scales, and a battery of cognitive and functional tests. This allowed us to examine the relationship between A $\beta$ , tau, and hypometabolism from A/T/N PET biomarkers. In turn, we aimed to uncover cognitive and functional impairments in the same individual to distinguish MCI from MD within AD [3, 22].

We assessed how combining  $^{18}\text{F}$ -Flutemetamol A-PET,  $^{18}\text{F}$ -MK-6240 T-PET and  $^{18}\text{F}$ -FDG N-PET, using literature-recommended cut-offs of the Centiloid, CenTauR, and AD metaROI standardised scales, described MCI and MD patients within the AD spectrum. We investigated the contribution of these three A/T/N PET biomarkers in a representative Norwegian clinical population of MCI and MD patients, randomised into two 2-tracer groups (Tau-A $\beta$  and Tau-FDG), to explore the diagnostic value of PET biomarkers that were semi-quantified with standardised scales. Specifically, we investigated: 1) whether binary A/T/N classification based on literature-recommended cut-offs differed between MCI or MD, hypothesising more positive A/T/N classifications in MD than MCI, 2) whether pathological burden semi-quantified from Centiloid A-PET, CenTauR T-PET, and AD metaROI N-PET, irrespective of being A/T/N (+/-), differed between MCI and MD, with the hypothesis of greater pathological burden in MD than MCI, and 3) assess the relationship between A-burden, T-burden, and N-burden with cognitive and functional tests across all participants, including interactions with diagnosis, hypothesising stronger correlations for tau and hypometabolism to cognition than for A $\beta$ .

## Materials and methods

### Participants

Individuals diagnosed by the referring clinician with MCI or MD due to AD according to the ICD-10 were recruited

from outpatient clinics at the Department of Geriatrics, Neurology, and Old Age Psychiatry at St. Olavs Hospital (Trondheim), Levanger Hospital (Levanger), Ålesund Hospital (Ålesund), and Haraldsplass Deaconess Hospital (Bergen). Exclusion criteria, in line with the 180°N consortium, included Magnetic Resonance Imaging (MRI) incompatibility (e.g., pacemaker), inability to provide consent or follow verbal instructions in the PET/MRI scanner, current or previous neurological and/or psychiatric diseases, and somatic diseases affecting participation.

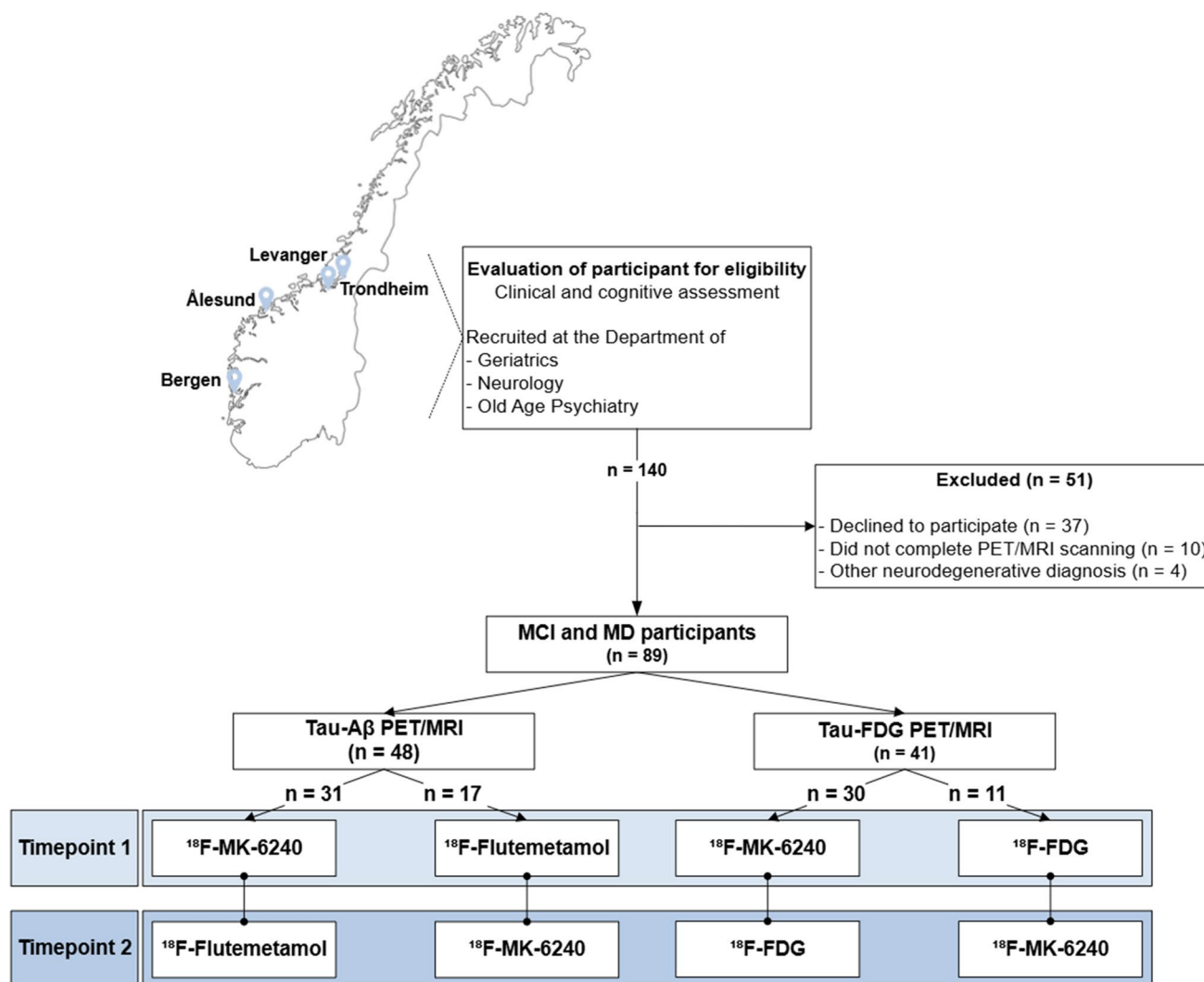
### Cognitive and clinical assessment

Participants underwent standard cognitive and clinical assessments in accordance with local guidelines (Fig. 1), and their relatives or caregivers were interviewed.

The following cognitive tests were performed: the Norwegian validated revised MMSE (MMSE-NR3), CDR, CERAD Word List Memory Test for immediate and delayed recall (CERAD) and recognition (CERAD R), Clock-drawing Test (CDT), Trail Making Test A (TMT-A) and B (TMT-B), Boston Naming Test (BNT), and the Controlled Oral Word Association Test (COWAT). Functional level was assessed with the Instrumental and Personal Activities of Daily Living (iADL, pADL) scale. Detailed descriptions and references of the cognitive, functional, and clinical assessments are included in the Supplementary Methods.

### PET/MRI

All PET and MRI were acquired on two Siemens Biograph mMR 3Tesla (T) PET/MRI scanners (Siemens Healthineers,



**Fig. 1** Flowchart of participant inclusion and scanning logistics for the  $^{18}\text{F}$ -MK-6240,  $^{18}\text{F}$ -Flutemetamol, and  $^{18}\text{F}$ -FDG PET/MRI protocols. Among the excluded participants, other neurodegenerative diagnoses ( $n=4$ ) included frontal lobe dementia, posterior cortical atrophy,

progressive speech apraxia and a general neurodegenerative condition. MCI: Mild Cognitive Impairment; MD: Mild Dementia; Aβ: Amyloid-β; PET: positron emission tomography; MRI: magnetic resonance imaging; FDG: fluorodeoxyglucose; n: number

Erlangen, Germany) with a 12-channel head coil, and conducted either at St. Olavs Hospital (Trondheim) or Haukeland Hospital (Bergen). The  $^{18}\text{F}$ -MK-6240 tracer was only available in Trondheim, whereas  $^{18}\text{F}$ -Flutemetamol and  $^{18}\text{F}$ -FDG were accessible in Trondheim and Bergen. Participants were randomised into two 2-tracer combination groups: the Tau-A $\beta$  group, scanned with  $^{18}\text{F}$ -Flutemetamol and  $^{18}\text{F}$ -MK-6240, and the Tau-FDG group, scanned with  $^{18}\text{F}$ -FDG and  $^{18}\text{F}$ -MK-6240 (Fig. 1). Tracer order was random with a mean (m) interval between  $^{18}\text{F}$ -Flutemetamol and  $^{18}\text{F}$ -MK-6240 scans of 64 days (standard deviation (SD)=80), and 71 days (SD=132) for  $^{18}\text{F}$ -FDG and  $^{18}\text{F}$ -MK-6240 scans (Fig. 1).

### PET acquisition and reconstruction

List mode PET data were acquired 90–120 min (m=83.9 and 88.4, SD=19.8 and 2.6) after intravenous injection of  $^{18}\text{F}$ -Flutemetamol (185 MBq (m=186.7, SD=5.9)) and  $^{18}\text{F}$ -MK-6240 (185 MBq (m=186.2, SD=8.1)). Prior to  $^{18}\text{F}$ -FDG examination, participants were instructed to fast for 6 h, drink 0.5 L of water, and their blood glucose levels were measured.  $^{18}\text{F}$ -FDG was administered intravenously (200 MBq (m=200.1, SD=5.1) if the glucose level was between 4.2–8.3 mmol/L. Participants laid down in a dark and noise-free room with a blindfold for 15 min before and 45 min after  $^{18}\text{F}$ -FDG injection. List mode PET data were acquired 60–80 min (m=57.9, SD=3.2) after injection [23]. Static PET images were reconstructed using the 90–110-min interval for  $^{18}\text{F}$ -Flutemetamol [24], 60–80 min for  $^{18}\text{F}$ -FDG [25], and 90–120 min for  $^{18}\text{F}$ -MK-6240 [26]. The reconstruction parameters are summarised in Table S1.

### 3T Brain MRI

This study used an ultrashort echo time (UTE) sequence for attenuation correction (AC), together with a 3D T1 MPRAGE, a 3D T2 SPACE and a 3D FLAIR. These sequences were used for semi-quantitative assessment of hippocampal atrophy according to the Medial Temporal Lobe Atrophy (MTA) score and white matter hyperintensity according to the Fazekas score [27, 28]. See Tables S2 and S3 for MRI scan parameters.

### Image processing

PMOD (version 4.405, PMOD Technologies LLC, Zürich, Switzerland) and Statistical Parametric Mapping (SPM) version 8 (<https://www.fil.ion.ucl.ac.uk/spm/software/spm8/>) were used for image processing. PET scans were coregistered with the T1-weighted MRI scans and spatially normalised to obtain the Standard Uptake Values (SUV) and

SUV ratios (SUVr) from the Centiloid, CenTauR, and AD metaROI scales, and whole-brain.

### $^{18}\text{F}$ -Flutemetamol A-PET Centiloid analysis

The  $^{18}\text{F}$ -Flutemetamol SUV was calculated from the PMOD PNEURO Centiloid pipeline and referenced to the whole cerebellum to obtain SUVr (SUVr<sub>cerebellum</sub>) values [11, 29], and directly converted into Centiloid values using the following equation [30]:

$$PMODCentiloid = (115.24 * SUVr_{cerebellum}) - 107.86$$

The cut-off for A+ was defined by Centiloid > 40 [29, 31]. Based on stopping rules in anti-amyloid treatment trials and current expert consensus, the Centiloid > 11, > 25, and > 30 cut-offs were used in sensitivity analyses [32].

### $^{18}\text{F}$ -MK-6240 T-PET CenTauR analysis

The  $^{18}\text{F}$ -MK-6240 SUV was calculated from the SPM8 CenTauR mask and referenced to the cerebellar cortex mask to obtain SUVr (SUVr<sub>cerebellum</sub>) values. Both masks were made available by the CenTauR Project (<https://www.gaain.org/centaur-project>). The  $^{18}\text{F}$ -MK-6240 SUVr<sub>cerebellum</sub> values were directly converted into CenTauR values using the following equation [12]:

$$SPM8CenTauR = (10.08 * SUVr_{cerebellum}) - 10.06$$

The cut-off for T+ was defined by CenTauR > 2 [12].

### $^{18}\text{F}$ -FDG N-PET AD metaROI analysis

The  $^{18}\text{F}$ -FDG SUV was calculated using the PMOD PNEURO maximum probability atlas (MPA) pipeline outlined with the AD metaROI mask and referenced to the pons, best to eliminate variability in tracer uptake and to distinguish between healthy and cognitively impaired individuals, to obtain SUVr (SUVr<sub>pons</sub>) values [14, 33]. The AD metaROI mask consisted of the bilateral angular gyrus, posterior cingulate cortex, and inferior temporal cortical area [34]. The cut-off for N+ was defined by AD metaROI < 1.41 SUVr<sub>pons</sub> [13, 14].

### A/T/N (+/-) classification

The  $^{18}\text{F}$ -MK-6240,  $^{18}\text{F}$ -Flutemetamol, and  $^{18}\text{F}$ -FDG PET scans were classified as either positive (+) or negative (-) for A ( $^{18}\text{F}$ -Flutemetamol A-PET), T ( $^{18}\text{F}$ -MK-6240 T-PET) and/or N ( $^{18}\text{F}$ -FDG N-PET) using the cut-off values specific

for each PET tracer in line with the A/T/N classification scheme [1]. We denote the cut-off-based classification as A+/-, T+/-, and N+/-.

### A-burden, T-burden, and N-burden

The total burden of A, T, and N was the continuous SUVR values derived from the Centiloid, CenTauR, and AD metaROI masks (Fig. 2). A higher A-burden and T-burden represent greater pathology, reflected by higher SUVR values, whereas a lower N-burden represents greater hypometabolism, reflected by lower SUVR values. In addition, whole-brain SUVR values were obtained from  $^{18}\text{F}$ -MK-6240,  $^{18}\text{F}$ -Flutemetamol, and  $^{18}\text{F}$ -FDG PET scans.

### Statistical analyses

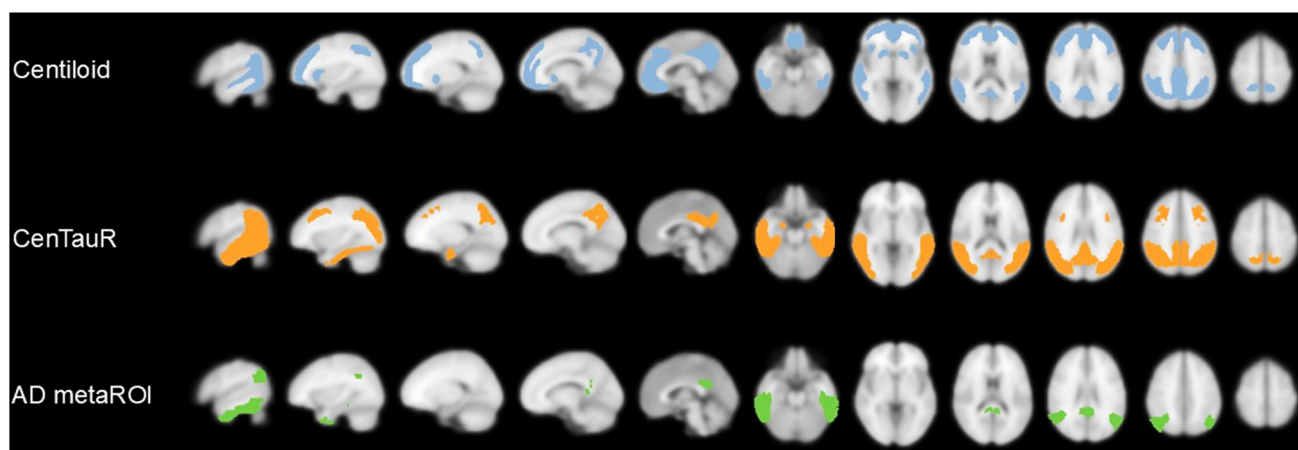
Demographic, cognitive, and functional differences between MCI and MD were assessed using Chi-Square tests for numerical variables and an independent sample t-test or Mann–Whitney U test for categorical variables. The frequency of A/T/N (+/-) classifications within the MCI and MD groups was calculated to investigate classification differences between diagnoses. Receiver operator characteristics (ROC) analyses assessed the diagnostic accuracy of A-burden, T-burden, N-burden, and whole-brain SUVR values in distinguishing MCI from MD. Optimal threshold values were defined using the top-left corner criterion. Differences in A-burden, T-burden, and N-burden between MCI and MD were then assessed with an independent sample t-test or Mann–Whitney U test to examine their relationship with diagnosis. Relationships between A-burden, T-burden, and N-burden and CSF A $\beta$ -42, pTau,

and tTau were assessed using Pearson's  $r$  and Spearman's  $\rho$  correlations. Differences in CSF A $\beta$ -42, pTau, and tTau between MCI and MD were assessed with an independent sample t-test. A descriptive analysis investigated the association between CSF A $\beta$ -42, tTau, and pTau and A/T/N (+/-) classifications and PET tracer burden. Multiple linear regression models were used with the cognitive scores as dependent variables, whereas logistic regression models were used with the functional scores as dependent variables. Models included A-burden, T-burden, and N-burden as main predictors, with age, sex, and education as covariates. Finally, we tested whether this relationship differed between MCI and MD by including diagnosis as an interaction term. RStudio (version 2025.05.0+496) was used for statistical analyses. Statistical significance was set at  $p < 0.05$ . No multiple-comparison adjustments were made due to the observational design.

## Results

### Participant characteristics

Table 1 shows the characteristics and group comparisons of MCI and MD. The MD participants were more cognitively impaired, with lower MMSE-NR3 ( $W = 1342.5$ ,  $p = 0.002$ ), lower CERAD ( $t(82) = 2.61$ ,  $p = 0.011$ ), and lower CERAD R ( $W = 1046.5$ ,  $p = 0.031$ ) scores, compared to the MCI group. The MD group experienced more functional impairments, as measured by the iADL scale ( $\chi^2(1) = 4.95$ ,  $p = 0.026$ ). For all other variables, MCI and MD did not differ significantly (Table 1 and S4), except for higher LDL cholesterol in the MD group ( $t(65) = -3.37$ ,  $p = 0.001$ ).



**Fig. 2** The Centiloid (A), CenTauR (T), and AD metaROI (N) masks overlaid on the MNI152 Magnetic Resonance Imaging template

**Table 1** Characteristics of all, MCI, and MD participants with statistical comparison between MCI and MD groups

|                                          | All ( <i>n</i> =89) | MCI ( <i>n</i> =45) | MD ( <i>n</i> =44) | <i>p</i> -value |
|------------------------------------------|---------------------|---------------------|--------------------|-----------------|
| <b>Sociodemographic variables</b>        |                     |                     |                    |                 |
| Age, years, mean (SD)                    | 70.5 (8.2)          | 71.0 (8.8)          | 70.1 (7.6)         | 0.430           |
| Sex, female/male, <i>n</i>               | 39/50               | 18/27               | 21/23              | 0.602           |
| Education, years, mean (SD)              | 13.9 (3.5)          | 13.7 (3.9)          | 14.1 (3.0)         | 0.593           |
| Marital status, married & co-living, (%) | 79.8%               | 75.6%               | 79.5%              | 1.000           |
| <b>Clinical assessment</b>               |                     |                     |                    |                 |
| <i>MRI biomarkers</i>                    |                     |                     |                    |                 |
| MTA score, right, median (IQR)           | 1 (1–3)             | 1 (1–2)             | 1 (1–3)            | 0.875           |
| MTA score, left, median (IQR)            | 2 (1–3)             | 2 (1–3)             | 2 (1–3)            | 0.192           |
| Fazekas score, median (IQR)              | 1 (1–2)             | 1 (1–2)             | 1 (1–2)            | 0.484           |
| <i>CSF biomarkers</i>                    |                     |                     |                    |                 |
| CSF Aβ-42 (ng/L), mean (SD)              | 709.2 (309.4)       | 730.6 (349.9)       | 685.7 (261.7)      | 0.651           |
| CSF tTau (ng/L), mean (SD)               | 290.9 (133.2)       | 272.7 (104.4)       | 310.7 (158.2)      | 0.331           |
| CSF pTau-181 (ng/L), mean (SD)           | 31.9 (18.3)         | 27.7 (12.4)         | 36.3 (22.3)        | 0.082           |
| <b>Cognitive tests</b>                   |                     |                     |                    |                 |
| MMSE-NR3 (0–30), mean (SD)               | 25.2 (3.3)          | 26.2 (2.9)          | 24.1 (3.5)         | <b>0.002</b>    |
| CDR (0–3), median (IQR)                  | 0.5 (0.5–0.8)       | 0.5 (0.5–0.5)       | 0.5 (0.5–1)        | 0.075           |
| CERAD (0–40), mean (SD)                  | 15.3 (5.9)          | 16.9 (5.7)          | 13.6 (5.7)         | <b>0.011</b>    |
| CERAD R (0–20), mean (SD)                | 15.8 (3.1)          | 16.4 (3.2)          | 15.2 (2.9)         | <b>0.031</b>    |
| CDT (0–5), median (IQR)                  | 5 (4–5)             | 5 (4–5)             | 4 (3–5)            | 0.121           |
| TMT-A, seconds, mean (SD)                | 54.7 (38.1)         | 52.1 (40.1)         | 57.4 (36.3)        | 0.468           |
| TMT-B, seconds, mean (SD)                | 130.3 (75.8)        | 125.9 (75.3)        | 134.7 (76.9)       | 0.345           |
| BNT (0–15), mean (SD)                    | 11.2 (2.4)          | 11.3 (2.6)          | 11.2 (2.2)         | 0.862           |
| COWAT, mean (SD)                         | 34 (14.6)           | 33.3 (15.7)         | 34.8 (13.5)        | 0.666           |
| <b>Functional tests</b>                  |                     |                     |                    |                 |
| pADL, impaired, (%)                      | 22.5%               | 13.3%               | 31.8%              | 0.067           |
| iADL, impaired, (%)                      | 46.1%               | 33.3%               | 59.1%              | <b>0.026</b>    |
| <b>Dementia risk factors</b>             |                     |                     |                    |                 |
| Diabetes, yes, (%)                       | 5.6%                | 8.9%                | 2.3%               | 0.371           |
| Hypertension, yes, (%)                   | 24.7%               | 31.1%               | 18.2%              | 0.243           |
| LDL cholesterol, mmol/L, mean (SD)       | 3.3 (1.3)           | 2.9 (1.0)           | 3.8 (1.4)          | <b>0.001</b>    |
| Serum-glucose, mmol/L, mean (SD)         | 5.7 (1.4)           | 5.9 (1.6)           | 5.6 (1.0)          | 0.334           |
| Creatinine, μmol/L, mean (SD)            | 77.6 (19.6)         | 80.5 (24.6)         | 74.6 (12.2)        | 0.443           |
| BMI, mean (SD)                           | 25.6 (3.6)          | 26.0 (3.0)          | 25.2 (4.2)         | 0.057           |
| <b>Smoking</b>                           |                     |                     |                    |                 |
| Current smokers, (%)                     | 11.2%               | 11.1%               | 11.4%              | 1.000           |
| Previous smokers, (%)                    | 31.5%               | 28.9%               | 34.1%              | 0.764           |
| Alcohol, units per week, mean (SD)       | 2.5 (2.5)           | 2.8 (3.0)           | 2.2 (2.0)          | 0.963           |
| Systolic blood pressure, mean (SD)       | 137.4 (19.6)        | 135.4 (18.9)        | 139.5 (20.4)       | 0.349           |
| Hearing impairment, yes, (%)             | 15.7%               | 22.2%               | 9.1%               | 0.159           |



**Table 1** (continued)

|                                          | All ( <i>n</i> =89) | MCI ( <i>n</i> =45) | MD ( <i>n</i> =44) | <i>p</i> -value |
|------------------------------------------|---------------------|---------------------|--------------------|-----------------|
| Visual impairment, yes, (%)              | 23.6%               | 26.7%               | 20.5%              | 0.660           |
| Other cardiovascular illnesses, yes, (%) | 30.3%               | 37.8%               | 22.7%              | 0.189           |
| Previous neurological incidents          |                     |                     |                    |                 |
| TBI, yes, (%)                            | 2.2%                | 0%                  | 4.5%               | 0.465           |
| Stroke, yes, (%)                         | 10.1%               | 8.9%                | 9.1%               | 0.972           |
| Other neurological illnesses, yes, (%)   | 4.5%                | 4.4%                | 4.5%               | 1.000           |
| Neuropsychiatric state                   |                     |                     |                    |                 |
| MADRS, mean (SD)                         | 3.8 (5.0)           | 4.2 (5.7)           | 3.5 (4.5)          | 0.886           |
| Previous depressive episodes, yes, (%)   | 11.2%               | 13.3%               | 9.1%               | 0.766           |
| Genetic                                  |                     |                     |                    |                 |
| ApoE4 allele status                      |                     |                     |                    |                 |
| No E4, (%)                               | 17 (25.8)           | 11 (34.4)           | 6 (17.6)           | 0.204           |
| One E4, (%)                              | 31 (47.0)           | 11 (34.4)           | 20 (58.8)          | 0.081           |
| Two E4, (%)                              | 18 (27.3)           | 10 (31.3)           | 8 (23.5)           | 0.669           |

*n* number; *SD* Standard deviation; *IQR* Interquartile range; *MCI* Mild Cognitive Impairment; *MD* Mild Dementia; *MRI* Magnetic Resonance Imaging; *MTA* Medial Temporal Lobe Atrophy; *CSF* Cerebrospinal fluid; *Aβ* Amyloid-β; *MMSE-NR3* Norwegian revised Mini-Mental State Examination; *CDR* Clinical Dementia Rating; *CERAD* CERAD Word List Memory Test for Immediate and Delayed recall; *CERAD R* CERAD Word List Memory Test for Recognition; *CDT* Clock-drawing Test; *TMT-A* Trail Making Test A; *TMT-B* Trail Making Test B; *BNT* Boston Naming Test; *COWAT* Controlled Oral Word Association Test; *pADL* Personal Activities of Daily Living; *iADL* Instrumental Activities of Daily Living; *LDL* Low-density Lipoprotein; *BMI* Body Mass Index; *TBI* Traumatic Brain Injury; *MADRS* Montgomery and Åsberg Depression Rating Scale; *ApoE*: Apolipoprotein E4

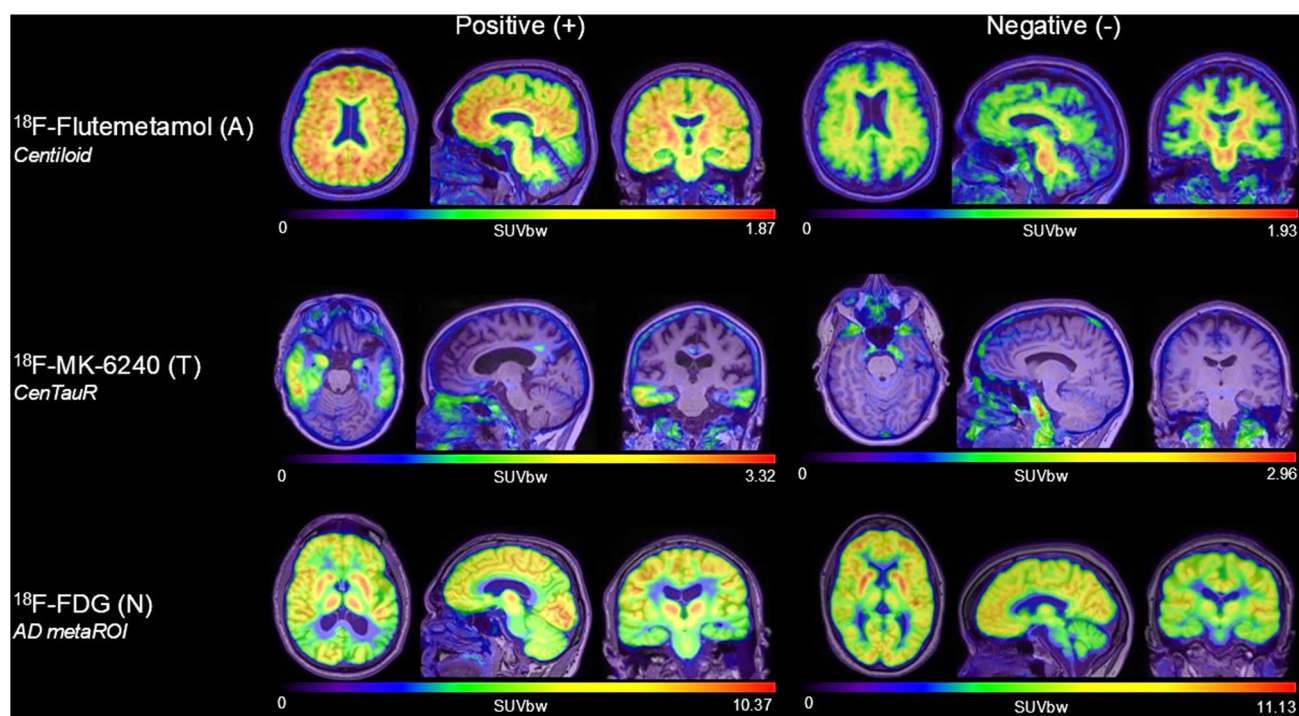
### A/T/N (+/-) classification and burden in the MCI and MD groups

Figure 3 visualises A-PET, T-PET, and N-PET scans classified within the A/T/N (+/-) scheme using Centiloid, CenTauR, and AD metaROI cut-offs. Being T+ was more frequent in the MD group (68%) compared to the MCI group (44%), regardless of the status of A or N (Fig. 4A and B). Both in the MCI and MD groups, being A+/T+ or N+/T+ were the most common classifications. However, more participants in the MD group were A+/T+ (39%) and N+/T+ (27%) compared to the MCI group (27% and 18%, respectively). 3% of the MD participants were T+ but A-. More participants in the MCI than the MD group were T- (56% versus 32%). The most frequent combinations were A+/T- or N+/T- in both the MCI and MD groups. The MCI group had a higher proportion of A+/T- (16% versus 9%) and N+/T- (31% versus 14%) than the MD group. In total, 9% of the MCI and 7% of the MD participants were both A- and T-, and 7% of the MD participants were both T- and N-. We observed that these A-/T-/N- participants had slightly higher CSF A $\beta$ -42 values (A-/T-:  $m = 1034.4$ ,  $SD = 556.9$ ; N-/T-:  $m = 513.0$ ) compared to individuals with a positive status for A, T, or N (Table S6). As for the sensitivity analysis of Centiloid cut-off

values, when the Centiloid cut-off was changed to  $>30$  instead of  $>40$ , only one individual in the T- MCI group moved to A+ (8.9% versus 6.7%) (Fig. S1). In the T- MD group, 3 (=6.8%) of the individuals moved from A+ to A- as the Centiloid cut-off changes from  $>11$  to  $>25$  and  $>25$  to  $>30$  (Fig. S2). Using continuous A-burden, T-burden, and N-burden values, T-burden was significantly higher in the MD group compared to the MCI group ( $W = 642$ ,  $p = 0.004$ ). No differences in A-burden or N-burden were observed between MCI and MD groups (Fig. 4C and S3, Table S5). Figure 5 and S5 and Table S7 show ROC curves and AUCs for A-burden, T-burden, N-burden and whole-brain ROIs SUVR to discriminate MCI from MD. Discriminability was poor ( $0.48 < \text{AUCs} < 0.68$ ).

### A/T/N-burden versus cognitive and functional scores

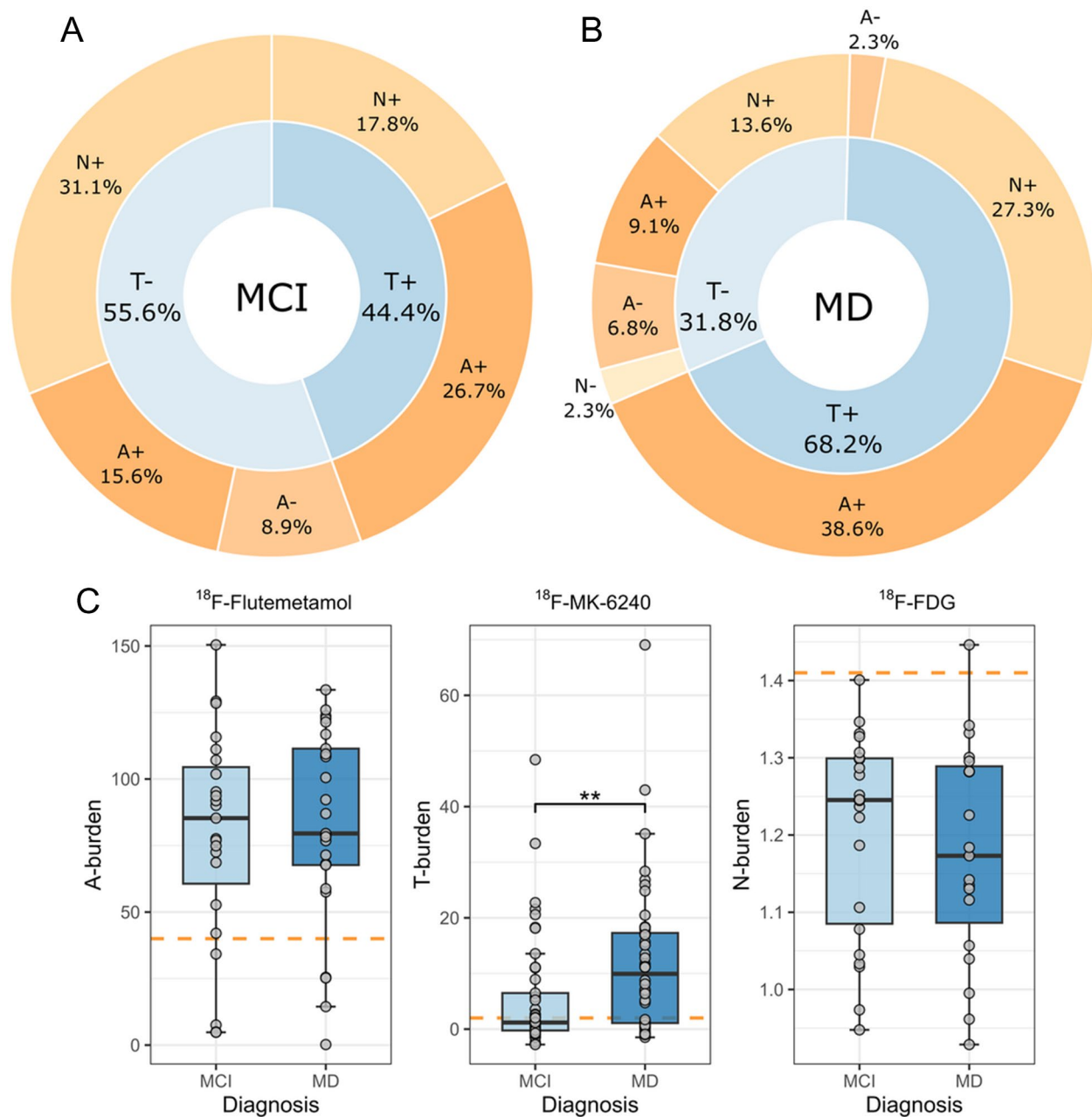
Table 2 shows significant results from the multiple linear regression on continuous cognitive scores. Figure 6 visualises these results alongside logistic regression results for the dichotomous functional scores. Higher A-burden, T-burden, and lower N-burden were significantly associated with lower MMSE-NR3 scores, higher A-burden and lower N-burden with lower CERAD scores, higher T-burden with lower CERAD R, and lower



**Fig. 3** Representative axial, sagittal, and coronal  $^{18}\text{F}$ -Flutemetamol (A),  $^{18}\text{F}$ -MK-6240 (T), and  $^{18}\text{F}$ -FDG (N) PET scans with SUVbw values of our participants. The images were classified as either posi-

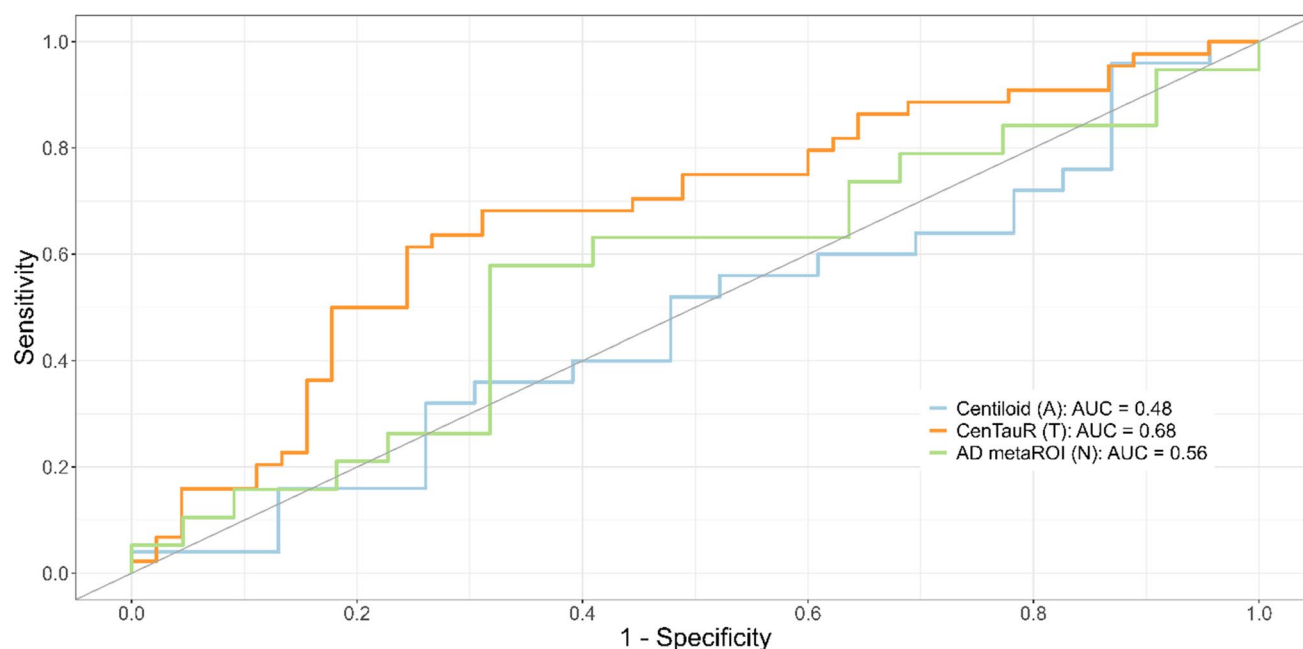
tive (+) or negative (-) for A, T, and N within the A/T/N classification scheme. SUVbw: Standard Uptake Value corrected for body weight





**Fig. 4** The upper row shows the distribution of A, T, and N (+/-) among those diagnosed with MCI (**A**) or MD (**B**). The blue ring indicates the percentage of those diagnosed with MCI (**A**) or MD (**B**) that have a positive or negative T-PET scan based on CenTauR > 2 threshold. The orange ring indicates the percentage of a positive or negative A-PET (Centiloid > 40) or N-PET (AD metaROI < 1.41) among those that are T (+/-). Panel C shows the A-burden, T-burden, and N-burden (median, range, outliers) with individual scatter points on the y-axis and diagnosis

group on the x-axis. A higher A-burden and T-burden represent greater pathology, reflected by higher SUVR values, whereas a lower N-burden represents greater hypometabolism, reflected by lower SUVR values. The orange lines indicate the cut-off level for the A/T/N (+/-) classification of each tracer. A: Amyloid- $\beta$ ; T: Tau; N: Neurodegeneration; MCI: Mild Cognitive Impairment; MD: Mild Dementia; \*\*:  $p < 0.01$



**Fig. 5** ROC curves indicating the diagnostic accuracy of A-burden, T-burden and N-burden that were derived from the Centiloid (A), CentTauR (T), and AD metaROI (N) scale to discriminate patients with

MCI from MD. AUC values for each scale are indicated in the plot. ROC: Receiver Operating Curve; AUC: Area Under the Curve

N-burden with lower CDT and COWAT scores (Fig. 6). After including an interaction term between A-burden, T-burden, or N-burden and diagnosis (MCI and MD), a lower N-burden was significantly associated with higher TMT-A scores in MD versus MCI (Fig. 7). The complete figure can be found as supplementary (Fig. S6).

## Discussion

To our knowledge, this is the first study to use a combination of A-PET, T-PET, and N-PET tracers with the Centiloid, CentTauR, and AD metaROI standardised scales, to study differences between MCI and MD patients in a population-based sample from several memory clinics.

In this multicentre observational study, we found that both A (+/-) and T (+/-) PET classification were most informative since N+ status, assessed by the AD metaROI, was positive in nearly all participants and therefore did not provide added value to differentiate MCI from MD. T-PET appeared to be a useful biomarker to distinguish MCI from MD at the group level, thereby outperforming A-PET and N-PET. This was not clearly reflected in our ROC analysis, possibly due to the small sample size. Although N-burden was correlated with most cognitive tests, T-burden was more strongly correlated with the cognitive tests and would have survived correction for multiple testing, highlighting the role of tau pathology in cognitive impairment. Moreover, the MMSE was

correlated with A-burden, T-burden, and N-burden, highlighting this test's sensitivity as a global measure of cognition and relationship to all the AD PET biomarkers. For the functional scores, no correlations were found with the A-burden, T-burden, and N-burden.

All in all, we found that T-PET appeared to aid diagnostic differentiation between MCI and MD and was most strongly correlated to cognitive impairment, while N-PET was more closely related to cognitive performance than to the clinical disease stage.

## The A/T/N classification, cut-off values, and clinical diagnosis

Our MCI and MD groups were similar in size and largely comparable, except that MD participants were, as expected, more cognitively and functionally impaired.

Using standard cut-offs, positive A/T or N/T (A+/T+ or N+/T+) classifications were common in both MCI and MD, but more so in MD. This aligns with studies using Centiloid and CentTauR [12, 17, 18] and the AD metaROI [35], reinforcing the A/T/N framework where A $\beta$  accumulation precedes tau, driving N along AD progression [3].

Focussing on T-PET, MCI participants were more frequently T- (56%), while the MD participants were more frequently T+ (68%). This matches findings of other studies implementing CentTauR and supports the idea that T pathology progresses over time in AD [3, 12, 17, 18]. Still, 32% of all MD participants were T-. This suggests that a CentTauR

**Table 2** Results of the multiple linear analyses with the cognitive scores as dependent variables. The models included A-burden, T-burden, and N-burden as main predictors, along with the interaction term with clinical diagnosis. Age, sex, and education were included as covariates. Only models with significant associations are presented, including information on both predictors and model-level statistics

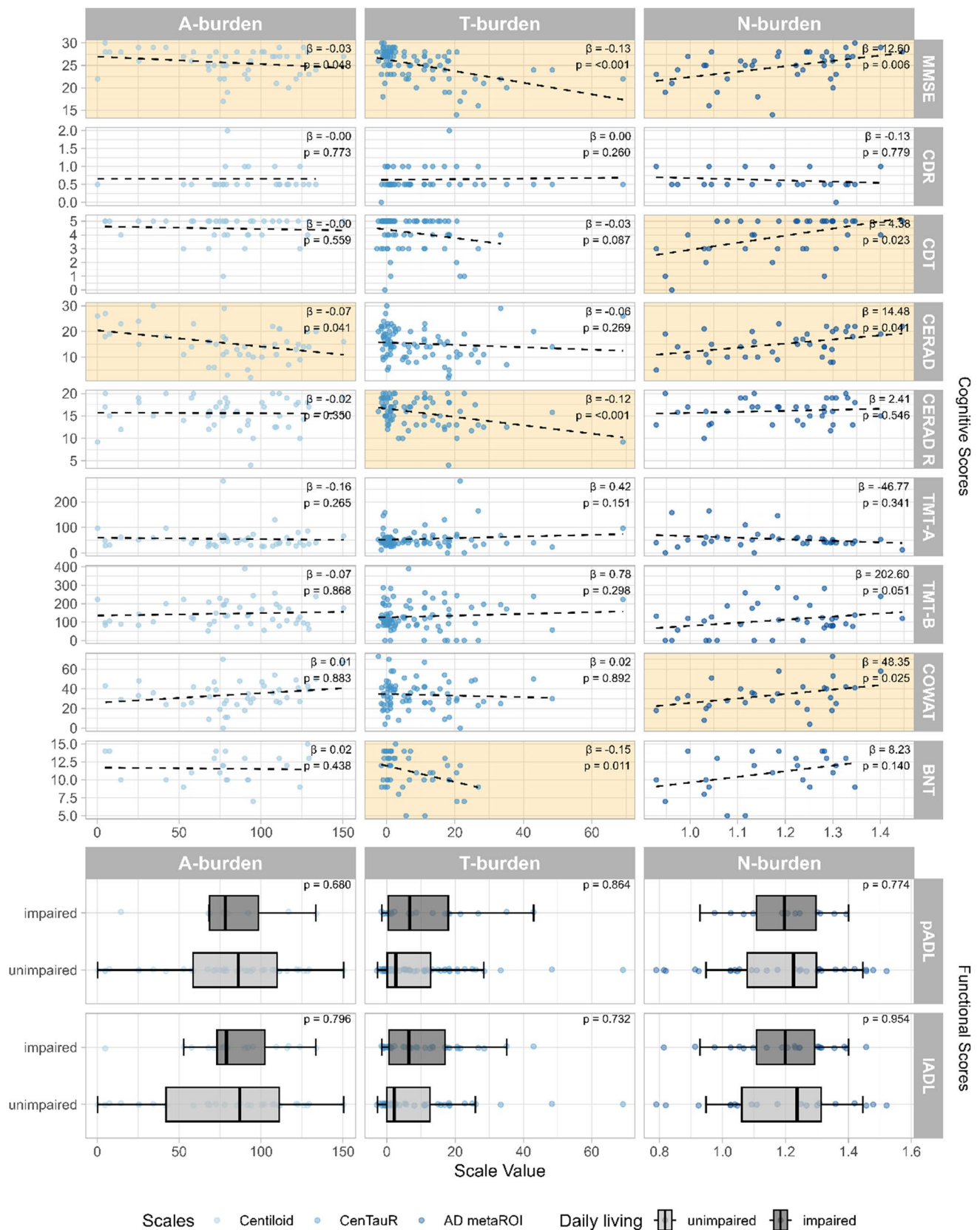
|                                    | $\beta$ | 95% CI        | p-value | F    | df1 | df2 | p-value          | adj. R <sup>2</sup> |
|------------------------------------|---------|---------------|---------|------|-----|-----|------------------|---------------------|
| <b>A-burden</b>                    |         |               |         |      |     |     |                  |                     |
| MMSE-NR3                           | -0.03   | -0.06, 0.00   | 0.048   | 2.94 | 4   | 30  | <b>0.037</b>     | 0.19                |
| CERAD                              | -0.07   | -0.14, 0.00   | 0.041   | 2.17 | 4   | 30  | 0.10             | 0.12                |
| <b>T-burden</b>                    |         |               |         |      |     |     |                  |                     |
| MMSE-NR3                           | -0.13   | -0.19, -0.08  | <0.001  | 7.95 | 4   | 65  | <b>&lt;0.001</b> | 0.29                |
| CERAD R                            | -0.12   | -0.18, -0.06  | <0.001  | 5.20 | 4   | 58  | <b>0.001</b>     | 0.21                |
| BNT                                | -0.15   | -0.26, -0.04  | 0.011   | 2.85 | 4   | 27  | <b>0.043</b>     | 0.19                |
| <b>N-burden</b>                    |         |               |         |      |     |     |                  |                     |
| MMSE-NR3                           | 12.60   | 3.82, 21.4    | 0.006   | 3.69 | 4   | 30  | <b>0.015</b>     | 0.24                |
| CDT                                | 4.38    | 0.66, 8.11    | 0.023   | 3.01 | 4   | 26  | <b>0.036</b>     | 0.21                |
| CERAD                              | 14.5    | 0.65, 28.31   | 0.041   | 1.88 | 4   | 26  | 0.14             | 0.10                |
| COWAT                              | 48.35   | 6.71, 89.98   | 0.025   | 2.67 | 4   | 20  | 0.06             | 0.22                |
| <b>N-burden * Diagnostic group</b> |         |               |         |      |     |     |                  |                     |
| TMT-A                              | -206.71 | -374.5, -38.9 | 0.018   | 3.83 | 6   | 26  | <b>0.007</b>     | 0.35                |

\* Interaction term; CI Confidence Interval; A Amyloid- $\beta$ ; T Tau; N Neurodegeneration; MMSE-NR3 Norwegian revised Mini-Mental State Examination; CERAD CERAD Word List Memory Test for Immediate and Delayed recall; CERAD R CERAD Word List Memory Test for Recognition; BNT Boston Naming Test; CDT Clock-drawing Test; COWAT Controlled Oral Word Association Test; TMT-A Trail Making Test A

threshold of 2 might be too strict for the MD group, as some individuals were classified as negative despite clinically relevant T-PET uptake in the CenTauR ROI. Our ROC analysis suggested that CenTauR > 2 may yield lower sensitivity and specificity compared with the original CenTauR paper [12]. However, while Villemagne et al. (2023) distinguished cognitively unimpaired A- from cognitively impaired A+ individuals, we aimed to distinguish MCI from MD, a more challenging comparison given their proximity on the AD spectrum. Lowering the CenTauR threshold to 1.5 or 1 appeared to increase the sensitivity slightly but reduced the specificity to MCI-MD differences. In contrast, increasing the threshold to 4.15 may preserve sensitivity while improving specificity relative to CenTauR > 2. Still, MD patients showed significantly higher T-burden than MCI, consistent with previous studies using CenTauR > 2 that distinguished MCI from MD [12, 17, 18].

Further, the T- MD group was classified as A+/T- (9%) and N+/T- (14%). The CSF analysis showed a pattern consistent with our A/T/N (+/-) PET classification (Table S6), where both A+/T- and N+/T- groups had pathological A $\beta$ -42 but normal tTau levels, whereas pTau was pathological only in the N+/T- group [36]. One possible explanation could be age and education. The T- MD was, on average, 6 years younger and had 2 more years of education than the A+/T+MCI group (*results not shown*). However, the N+/T+MCI group had one more year of education than the N+/T- MD group (*results not shown*). Thus, education alone did not explain the findings under the cognitive reserve (CR) theory, which posits that education improves CR, thereby reducing vulnerability to cognitive impairment despite comparable neuropathology [37]. Another explanation is that coexisting pathologies in the T- MD group may drive A $\beta$  and neurodegeneration, causing symptoms independent of T pathology [3].

For A-PET, there were participants in both the MCI (9%) and MD (7%) groups, who were classified with A-/T-. In addition, 3% of MD participants were A-/T+. This finding is unexpected, as AD progression is thought to start with A $\beta$  accumulation [3]. The supplementary CSF analysis disagreed with our A/T/N (+/-) PET classification, as both A-/T+ and A+/T+ showed pathological A $\beta$ -42, whereas tTau was elevated only in A+/T+ [36]. This may reflect age, as the A- MD was on average 16 and 18 years younger than the A+/T+MD and MCI groups, respectively (*results not shown*). Further, co-pathologies and misdiagnosis cannot be ruled out [3]. It could also imply that the cut-off level for A is too strict, as one would expect all MD participants to be A+. We used the Centiloid > 40 cut-off for A+ status, which is most concordant with visual reads [29, 31], but others have applied cut-offs between 10–30 Centiloid [32]. Given the ongoing debate about the optimal Centiloid threshold





**Fig. 6** The associations between the cognitive and functional scores on the y-axis and A-burden, T-burden, and N-burden on the x-axis, with significant associations highlighted in yellow. A higher A-burden and T-burden represents greater pathology, reflected by higher SUVR values, whereas a lower N-burden represents greater hypometabolism, reflected by lower SUVR values A: Amyloid- $\beta$ ; T: Tau; N: Neurodegeneration; MMSE: Norwegian revised Mini-Mental State Examination; CDR: Clinical Dementia Rating; CDT: Clock-drawing Test; CERAD: CERAD Word List Memory Test for Immediate and Delayed recall; CERAD R: CERAD Word List Memory Test for Recognition; TMT-A: Trail Making Test A; TMT-B: Trail Making Test B; COWAT: Controlled Oral Word Association Test; BNT: Boston Naming Test; pADL: Personal Activities of Daily Living; iADL: Instrumental Activities of Daily Living; MCI: Mild Cognitive Impairment; MD: Mild Dementia

depending on its use (therapy stopping versus inclusion), we performed a sensitivity analysis with commonly used cut-offs of  $>11$ ,  $>25$ , and  $>30$  to see if these changed the proportion of A (+/-) individuals [14, 29, 31, 32]. Applying these alternative cut-offs only changed a minority of A+ and T- individuals but did not substantially shift the number of individuals with A (+/-) proportions in the MCI or MD groups (Fig. S1 and S2). In addition, a Centiloid  $>40$  reliably detected A $\beta$  pathology [32] and maximises sensitivity and specificity using only A-PET without CSF or postmortem analyses [29, 31]. Therefore, using A-PET with a Centiloid  $>40$  classifies most of our sample, captures A/T (+/-) differences between MCI and MD, and appears to provide consistent results across studies.

Differences in A+/- status between the MCI and MD groups were not reflected in A-burden levels. A-burden levels overlapped and were highly variable in both groups, suggesting substantial individual differences in the MCI and MD groups. Some studies found significant A-burden differences between MCI and MD using Centiloid cut-offs of 25 and 30 [19, 38], while others observed differences at 20 and 21 [18, 39]. Our results do not support such a distinction (Fig. S3). The main difference between these studies is the definition of the dementia groups. Studies reporting significant A-burden differences included mixed-aetiology dementia groups (e.g., frontotemporal lobe dementia), while those without significant differences included AD-spectrum participants concurring with our results. Since our MD group is AD-spectrum based on clinical evaluation, this may explain the absence of significant A-burden differences between MCI and MD, highlighting the importance of study cohort differences and participant selection.

N- was only seen in the MD group (7%). This outcome might also reflect the SUVR  $<1.41$  cut-off for N+, as almost everyone (97%) was N+ at this threshold. According to the original papers, this cut-off value is conservative, providing the best trade-off between sensitivity, specificity, and accuracy for distinguishing MCI from MD [13, 14]. Although

our results align with Jack et al. (2017), our ROC analysis suggest that AD metaROI  $<1.41$  may yield high sensitivity but minimal specificity, whereas  $<1.19$  may provide a favourable trade-off for distinguishing MCI from MD with N-PET in our population. This could explain why almost all our participants were classified as N+, possibly incorrectly. Jack et al. (2017) acknowledge that no N (+/-) cut-off reliably indicates the onset of pathological hypometabolism, especially for distinguishing MCI from MD within the AD spectrum [14]. Without a clear AD metaROI threshold, N-PET's utility to distinguish MCI from MD appears limited.

Altogether, our analysis of MCI and MD participants scanned with A-PET, T-PET, and N-PET tracers showed that A+ and T+ were most common in both groups. T+ was more frequent in the MD group, while N+ was nonspecific. T-PET appears to be a potentially useful biomarker for distinguishing MCI from MD based on T-burden, as A-burden and N-burden were similar across groups.

### A-burden, T-burden, and N-burden biomarkers, cognition, and function

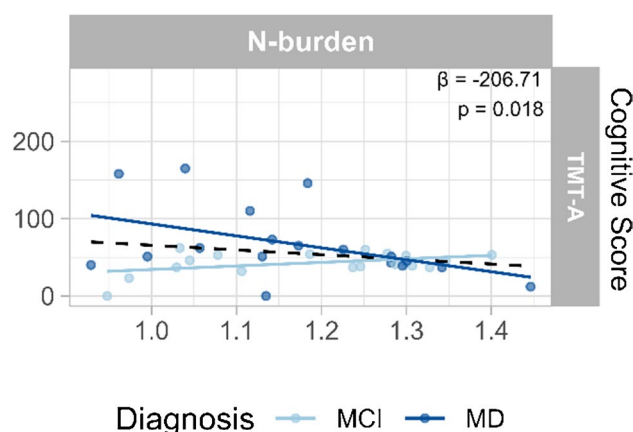
Our last analysis included a regression analysis to investigate the correlation between A/T/N burden and cognitive and functional scores across all participants and their interaction with diagnosis.

Across A-PET, T-PET, and N-PET, the N-burden was associated with most cognitive tests, followed by T-burden and A-burden. This finding is consistent with previous studies, with hypometabolism showing the strongest association, followed by tau, while A $\beta$  burden appears least related to cognition [13, 20, 21], adding to our understanding of AD pathology.

Turning to the specific cognitive tests, a higher A-burden ( $\beta = -0.03$ ,  $p = 0.048$ ) and T-burden ( $\beta = -0.13$ ,  $p < 0.001$ ) and lower N-burden ( $\beta = 12.6$ ,  $p = 0.006$ ) were significantly correlated with lower MMSE scores, a measure of general cognitive functioning. We observed a more significant relationship between MMSE and T-burden than with A-burden, which is in line with the general finding that T-PET is more strongly associated with cognition than A-PET [13, 16]. One study that used both Centiloid for A-burden and Cent-TauR for T-burden found similar results to our study [19]. Similarly, N-burden correlated significantly with MMSE in our study, in line with studies implementing the same AD metaROI as well as other ROIs [13, 19, 21].

Lower scores on the memory recognition test CERAD R and the naming BNT tests were significantly correlated with higher T-burden. Previous studies correlating cognitive tests with T-burden from PET scans, analysed with various





**Fig. 7** The association between the cognitive score TMT-A on the y-axis and N-burden on the x-axis with diagnoses (MCI in light blue versus MD in dark blue) as an interaction term. A lower N-burden represents greater hypometabolism, reflected by lower SUVR values. The black striped line shows the association between N-burden and TMT-A, excluding the interaction term. N: Neurodegeneration; TMT-A: Trail Making Test A; MCI: Mild Cognitive Impairment; MD: Mild Dementia

methods, or CSF reported similar findings [17, 20]. Next, a higher A-burden and lower N-burden significantly correlated with lower scores on the memory test CERAD. Previous studies also reported significant negative correlations between CERAD, or a similar memory test, and Centiloid or SUVR from N-PET [20, 40, 41]. For N-burden, higher scores on the spatial memory CDT and verbal COWAT tests were significantly correlated with lower N-burden. Although limited, a few studies suggest that hypometabolism leads to verbal and spatial impairments due to synaptic loss in the parietal and frontal lobes in AD [42–44]. After including diagnosis (MCI or MD) as an interaction term in the analyses, only N-burden was significant, underscoring  $^{18}\text{F}$ -FDG N-PET as a sensitive biomarker for evaluating brain function in AD [8].

We found no significant relationship between A-burden, T-burden, and N-burden and the functional iADL and pADL tests. Since the loss of functionality and its independence is what differentiates MCI from MD in the clinic [2], we expected different relationships between diagnosis and A/T/N burden [45, 46]. A limitation of our iADL and pADL measures was the use of dichotomous variables instead of the full score range, as well as the use of different test versions across clinics. This discrepancy could explain why we have not found any of the previously described relationships [45].

Overall, our regression analyses revealed that progressive worsening of cognitive scores was present with increasing A-burden, T-burden, and N-burden, where N-burden was associated with most cognitive tests, followed by T-burden and A-burden.

## Limitations

Limitations of the study include that all three radiotracers should have been included per participant, however, clinicians and the Regional Committees for Medical and Health Research Ethics for the region of mid-Norway (REK) deemed two tracers as the limit due to radiation exposure. No inclusion and demographical bias was present due to Tau-A $\beta$  and Tau-FDG grouping. We implemented literature-recommended Centiloid, CenTauR, and AD metaROI cut-offs to assess the A/T/N classification in MCI and MD participants [12–14, 29, 31, 32]. Their novelty complicates result interpretation. Moreover, our clinical sample also included imaging biomarker-negative individuals. The level of the cut-offs is therefore crucial, as these individuals are still experiencing cognitive deficits and co-pathologies are common. Lastly, we tested multiple biomarkers in a relatively small sample, which may have reduced statistical power and warrants cautious interpretation of our results. If adjusted for multiple comparisons, T-burden would show stronger correlations, although N-burden would still be associated with more cognitive tests.

## Conclusion

This multicentre observational study demonstrates that the A/T/N (+/-) classifications, using standardised A/T/N PET scales (Centiloid, CentauR, and AD metaROI) can identify differences between MCI and MD patients, with T-PET appearing to be a useful biomarker to distinguish MCI from MD. N-PET was associated with most cognitive tests, while T-PET was more strongly correlated. Given the novelty of the standardised scales for A/T/N PET biomarkers, future validation is required to ensure a robust and reproducible workflow. Standardising A/T/N biomarkers may enable cross-study neuroimaging comparisons, thereby strengthening our understanding of the AD spectrum, improving diagnosis, and future therapeutic research.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s00259-026-07911-5>.

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**Author contributions** LE, AKH, SA, RSE and ITS contributed to the study conception and design. Patient recruitment and data collection were done by EMB, PWS, SA, NB, RSE, JCJF, GRG, KH, GTL, SHKM, SRM, PM, LTM, YMS, and RES. Data analysis was performed by VRM. The first draft of the manuscript was written by VRM, where LE, AKH, KVL, and EGM initially commented on previous versions of the manuscript. All authors read, revised, and approved the final manuscript.

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**Data availability** The datasets generated during the current study are available upon reasonable request to the corresponding author and after complete finalisation of the study, also taking all needed ethical and possible contractual approvals into consideration.

## Declarations

**Ethical approval** Ethical approval for this study was obtained from the Regional Committees for Medical and Health Research Ethics for the region of mid-Norway (REK number 536745).

**Informed consent** All participants provided written informed consent before any study-related procedures were conducted.

**Competing interests** SA has been a consultant in Eisai's National Norwegian Alzheimer's disease Advisory Board meetings.

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